

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CHEMISTRY

9729/03

Paper 3 Free Response

10 September 2018

2 hours

Candidates answer on separate paper.

Additional Materials: Writing Papers
Data Booklet
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

You are advised to show all working in calculations.
You are reminded of the need for good English and clear presentation in your answers.
You are reminded of the need for good handwriting.
Your final answers should be in 3 significant figures.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part question.

This document consists of xx printed pages.



Section A

Answer **all** the questions in this section.

1 This question is about the chemistry of ethene and its derivatives.

(a) Ethene is the starting material to form ethanedioic acid.

(i) Suggest the synthetic route for the formation of ethanedioic acid from ethene. [2]



Step 1: cold, KMnO_4 , NaOH(aq) [✓]

Step 2: $\text{K}_2\text{Cr}_2\text{O}_7$, dil H_2SO_4 , heat [✓]

(ii) 0.200 mol of ethene is stored in a 20.0 dm^3 flask with 0.800 mol of ethane at 127°C .

Calculate the total pressure in the flask. Hence or otherwise, calculate the partial pressure of ethene in the flask. [2]

$$P(20.0 \times 10^{-3}) = (0.200 + 0.800)(8.31)(127 + 273)$$

$$P = 166200 \text{ Pa} \text{ [1]}$$

$$\text{Partial pressure of ethene} = 0.2/1.0 \times 166200 = 33240$$

$$= 33.2 \times 10^3 \text{ Pa} \text{ [1]}$$

(iii) The total pressure that you calculated in (a)(ii) is different from the actual pressure exerted. Suggest an explanation for the difference. [1]

There is significant intermolecular forces of attraction present between the gases OR intermolecular forces of attraction between the gases are not negligible [1].

(iv) Ethene reacts with hydrogen in the presence of nickel catalyst to form ethane.

Explain why nickel can be used as a catalyst in this reaction. [2]

Nickel has **available 3d electrons for bond formation** with the reactant molecules (ie. ethene and hydrogen) [1] and **available low lying vacant orbitals or energetically accessible orbitals which can accept electron pairs** from the reactant molecules. [1]

(v) Outline the mode of action of the catalyst in this reaction. [2]

The catalyst is in a solid state and it functions as a heterogeneous catalyst [✓] as it is in a different phase than ethene and hydrogen. Ethene and hydrogen (reactants) will be adsorbed to the surface of the Ni catalyst. [✓] Bonds in the reactant molecules are weakened [✓] which lowers the activation energy. New bonds are then formed between adjacent reactant molecules to form ethane (product). The product formed will be desorbed from the surface of the Ni catalyst. [✓]

(b) Dissolving 4.82×10^{-5} mol calcium ethanedioate, CaC_2O_4 , in 1 dm^3 of water forms a saturated solution.

(i) Write an expression for the solubility product of calcium ethanedioate and state its units. [1]

$$K_{sp}(\text{CaC}_2\text{O}_4) = [\text{Ca}^{2+}][\text{C}_2\text{O}_4^{2-}] \quad \text{unit: mol}^2 \text{ dm}^{-6} \quad [1]$$

- (ii) Calculate the solubility product of calcium ethanedioate. [1]

$$K_{sp} = (4.82 \times 10^{-5})^2 \\ = 2.32 \times 10^{-9} \quad [1]$$

[units not nec. as already stated in (i)]

- (iii) 50.0 cm³ of 0.100 mol dm⁻³ CaCl₂ and 50.0 cm³ of 0.300 mol dm⁻³ Na₂C₂O₄ are mixed together. Determine if CaC₂O₄ will be precipitated out. [2]

For precipitation to occur, IP of CaC₂O₄ must exceed K_{sp} of CaC₂O₄
Since the volume used is the same, the new concentration is halved the original concentration.

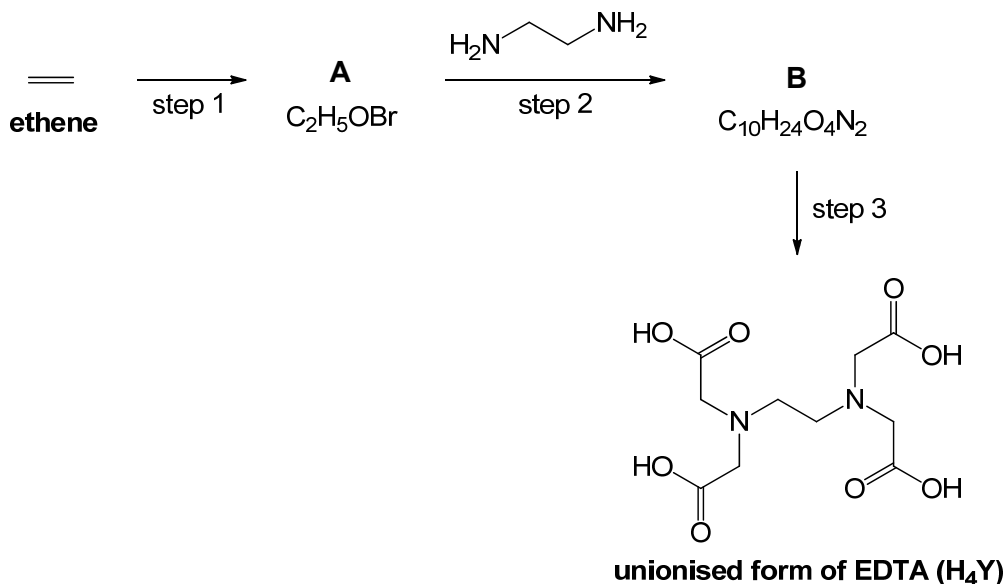
$$\text{New } [\text{Ca}^{2+}] = 0.100 / 2 = 0.05 \quad [\checkmark]$$

$$\text{New } [\text{C}_2\text{O}_4^{2-}] = 0.300 / 2 = 0.15 \quad [\checkmark]$$

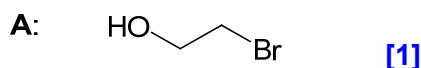
$$\text{IP of CaC}_2\text{O}_4 = (0.05)(0.15) = 7.50 \times 10^{-3} \quad [\checkmark]$$

Since IP > K_{sp}, CaC₂O₄ will ppt out. [✓][ECF with student's K_{sp} value from (ii)]

- (c) Synthesis of the unionised form of EDTA (H₄Y) can be carried out in the laboratory using ethene as the starting material.

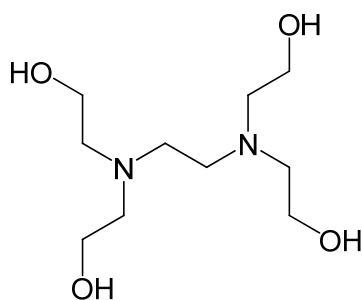


- (i) Draw the structures of **A** and **B**. [2]



4

B:



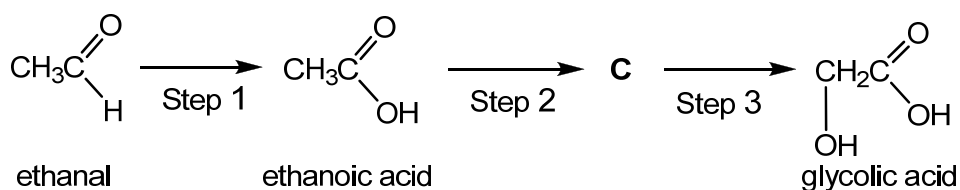
[1]

- (ii) Suggest appropriate reagents and conditions for step 3.

[1]

Step 3: dilute H_2SO_4 , $\text{KMnO}_4(\text{aq})/\text{K}_2\text{Cr}_2\text{O}_7(\text{aq})$, heat under reflux [1] (followed by careful addition of $\text{NaOH}(\text{aq})$)

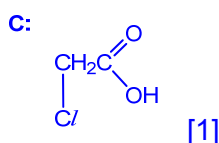
- (d) Ethene can be used to form ethanal. Ethanal in turn is used to synthesize glycolic acid via the reaction shown below.



- (i) Identify the intermediate **C** and state the reagent and condition for Step 3.

[2]

Step 3: $\text{NaOH}(\text{aq})$, heat under reflux; followed by aq. HCl [1]



- (ii) Explain the difference in acidity between glycolic acid and ethanoic acid.

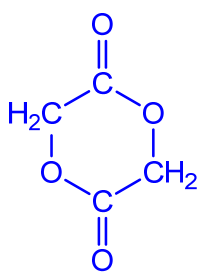
[2]

In glycolic acid, the electron withdrawing $-\text{OH}$ [✓] group reduces the negative charge on $-\text{COO}^-$ [✓] of the conjugate base of glycolic acid, thus making the anion of glycolic acid is more stable than the ethanoate anion [✓]. Hence, glycolic acid is a stronger acid [✓].

- (iii) Two molecules of glycolic acid can react with one another under suitable conditions to form a neutral compound with the loss of two water molecules.

Suggest a possible structure for the compound formed.

[1]



- (iv) Suggest a simple chemical test to distinguish between ethanoic acid and glycolic acid.

[2]

Test: $\text{K}_2\text{Cr}_2\text{O}_7$, dil H_2SO_4 , heat [1]

Ethanoic acid: orange $\text{K}_2\text{Cr}_2\text{O}_7$ turns green [✓]

Glycolic acid: orange $\text{K}_2\text{Cr}_2\text{O}_7$ remains [✓]

OR

Test: KMnO_4 , dil H_2SO_4 , heat

Ethanoic acid: purple KMnO_4 remains

Glycolic acid: purple KMnO_4 decolorised. Effervescence observed. gas produced forms white ppt with $\text{Ca}(\text{OH})_2$

[Total: 23]

- 2 Iron is the fourth most common element in the Earth's crust. It is a d-block element which is known to exhibit different characteristics from the s-block elements. Since ancient times, iron has been widely employed in a variety of applications.

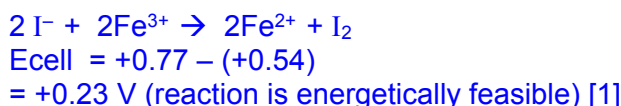
- (a) (i) One well-known property of iron and its compounds is the ability to catalyse reactions. For example, aqueous iron(II) chloride can be used to catalyse the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$, to form I_2 and SO_4^{2-} ions.

Using relevant $E^\ominus$ values from the *Data Booklet*, explain why iron(II) chloride can be used as a catalyst for this reaction.

[2]



$$\begin{aligned} E_{\text{cell}} &= +2.01 - (0.77) \\ &= +1.24 \text{ V (reaction is energetically feasible) [1]} \end{aligned}$$



- (ii) A sample of iron was vapourised, ionised and passed through an electric field. Analysis of deflection occurring at the electric field region revealed that a sample of $^{32}\text{S}^{2-}$ ions would be deflected by $+20^\circ$ towards the positive potential.

What is the angle, and direction of deflection for a sample of $^{56}\text{Fe}^{3+}$ ions passing through the same electric field?

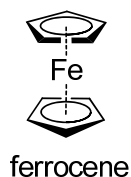
[2]

$$\begin{aligned} \text{Angle of deflection of } ^{32}\text{S}^{2-} &= 20 \\ 20 &= k (-2 / 32) \\ k &= -320 \end{aligned}$$

$$\begin{aligned} \text{Angle of deflection of } ^{56}\text{Fe}^{3+} &= (-320) (3 / 56) \\ &= -17.1^\circ \quad [1 \text{ for final answer; } 1\text{m for indication of direction}] \end{aligned}$$

or 17.1° [1m] towards the negative potential [1m]

- (b) Ferrocene, $\text{Fe}(\text{C}_5\text{H}_5)_2$, is an orange solid which is known to exhibit anti-cancer activity. In this complex, C_5H_5^- is the ligand and it donates π electrons from the ring to the vacant 3d orbital of Fe. The structure of ferrocene is given below.



- (i) State the oxidation number of Fe in ferrocene. [1]
+2

- (ii) Suggest why ferrocene is a coloured complex. [2]
There is partially filled 3d orbitals in Fe^{2+} . [✓]

In the presence of ligands, 3d orbitals of iron are split into 2 groups with small energy gap (d–d splitting). [✓]

Some light energy is used to promote an electron [✓] from a d-orbital of lower energy to a d-orbital of a higher energy [✓].

(Colour observed is complementary to the colour absorbed.)

- (iii) Light of a longer wavelength is lower in energy than light of a shorter wavelength. The following table shows the spectral colours and the corresponding wavelengths.

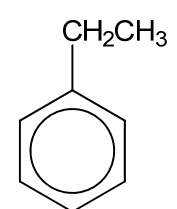
Colour	Wavelength / nm
Violet	380 – 450
Blue	450 – 495
Green	495 – 570
Yellow	570 – 590
Orange	590 – 620
Red	620 – 750

Given that aqueous Fe^{2+} ion is green in colour, suggest and explain if water causes a larger split between the two groups of 3d orbitals as compared to C_5H_5^- . [2]

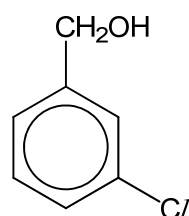
Water causes a smaller split between the 3d orbitals. [1]

With water as the ligand, electromagnetic waves with energy corresponding to red is absorbed which has a lower energy than blue, which is absorbed when C_5H_5^- is the ligand. [1]

- (c) Ethylbenzene is used to synthesize (3-chlorophenyl)methanol which is used as a general solvent for inks, paints, lacquers, epoxy resin coatings and as a degreasing agent.



ethylbenzene

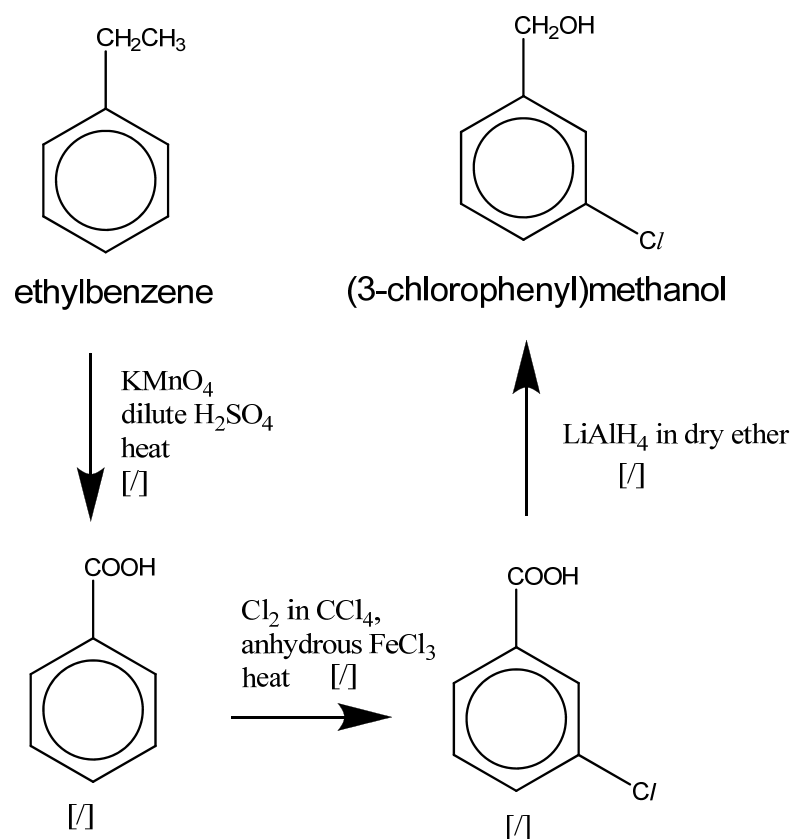


(3-chlorophenyl)methanol

Starting with ethylbenzene, outline a three-step reaction scheme to obtain (3-chlorophenyl)methanol. Your answer should include clearly the reagents and conditions in each step, and the structures of all intermediates formed.

[3]

Ans:



5 [✓] – 3m
 3 to 4 [✓] – 2m
 2 [✓] – 1m

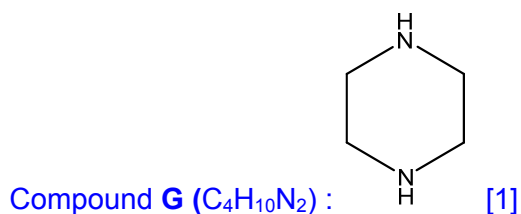
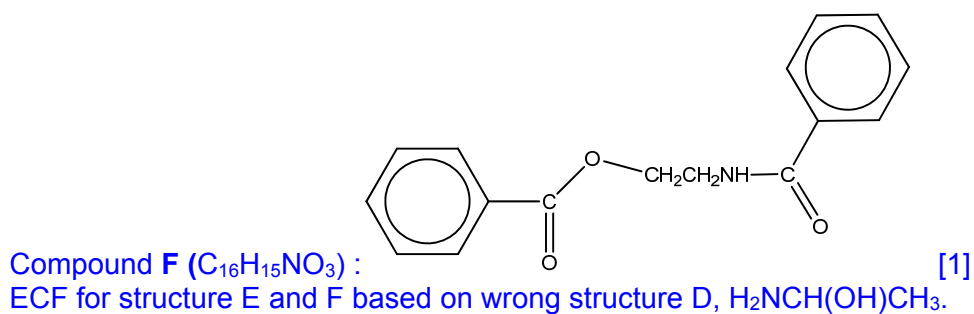
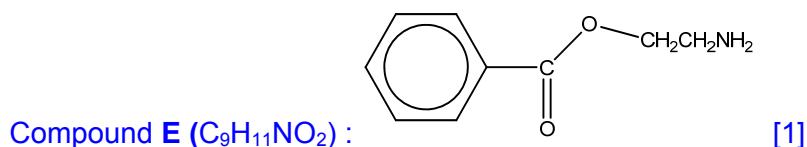
- (d) **D** is an achiral organic compound with the molecular formula $\text{C}_2\text{H}_7\text{NO}$. It can be formed from the reaction between a primary amide and lithium aluminium hydride. In the presence of a suitable catalyst, 1 mole of **D** reacts with 1 mole of benzoic acid to form **E**, $\text{C}_9\text{H}_{11}\text{NO}_2$. However, 1 mole of **D** requires 2 moles of benzoyl chloride to react completely to form **F**, $\text{C}_{16}\text{H}_{15}\text{NO}_3$ and copious white fumes. 2 moles of **D** can also react with gaseous PCl_5 to form a cyclic **G**, $\text{C}_4\text{H}_{10}\text{N}_2$, which contains a 6-membered ring.

Deduce the structures of **D**, **E**, **F** and **G** and explain the reactions described.

[7]

Info	Deduction
D is formed from reaction between primary amide and LiAlH_4	D contains a <u>primary amine</u> [✓]
1 mol D reacts with 1 mol of benzoic acid to form E , $\text{C}_9\text{H}_{11}\text{NO}_2$	D undergoes <u>condensation</u> / nucleophilic acyl substitution [✓] E is an <u>ester</u> [✓] D contains <u>alcohol</u> [✓]
1 mol D require 2 mol benzoyl chloride to react completely to form Compound F , $\text{C}_{16}\text{H}_{15}\text{NO}_3$ and copious white fumes.	D undergoes <u>condensation</u>/ nucleophilic acyl substitution [✓] - Copious white fumes are <u>HCl</u> [✓] F contains an <u>amide</u> [✓] and an <u>ester</u> [✓]
2 mol D reacts with gaseous PCl_5 to form a cyclic Compound G , $\text{C}_4\text{H}_{10}\text{N}_2$	D undergoes <u>nucleophilic substitution</u> [✓] G contains an <u>amine</u> [✓]
8 - 10 [✓] – 3m 5 - 7 [✓] – 2m 2 – 4 [✓] – 1m	

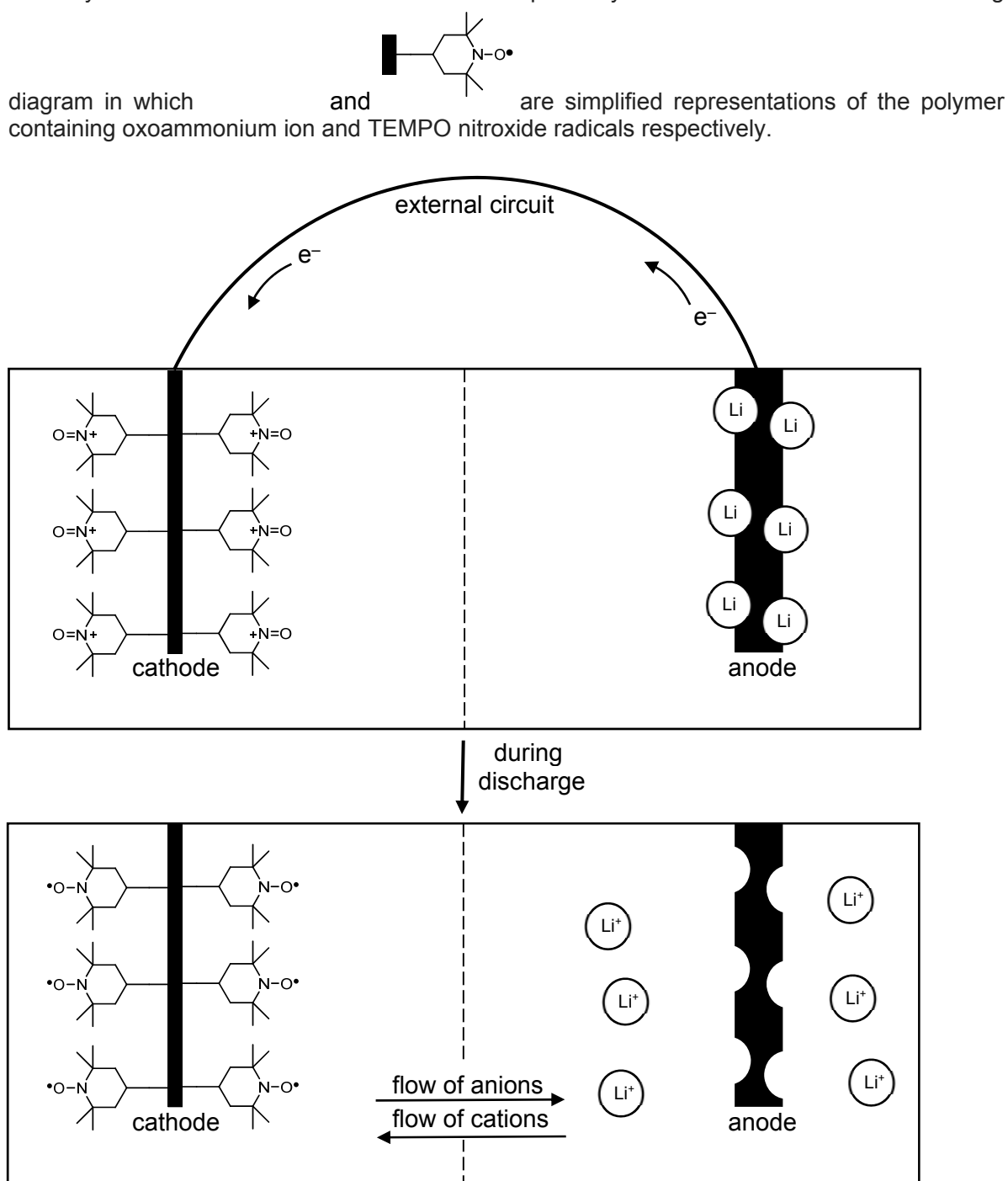
D ($\text{C}_2\text{H}_7\text{NO}$) : $\text{NH}_2\text{CH}_2\text{CH}_2\text{OH}$ [1]



[Total: 19]

- 3 An organic radical battery (ORB) is a relatively new type of battery which uses flexible plastics, to provide electrical power. One type of hybrid ORB/Li-ion battery consists of: a cathode made from solid organic polymer containing oxoammonium ion formed from (2,2,6,6-Tetramethylpiperidin-1-yl)oxyl (TEMPO) nitroxide radicals; an anode made from graphite with Li atoms inserted between the graphite layers; and an electrolyte of LiPF_6 dissolved in organic solvent.

During discharge, Li atoms give up electrons at the anode to become Li^+ ions. The electrons travel round the external circuit, and are picked up by the cathode. The anions and cations in the electrolyte move to the anode and cathode respectively. This is illustrated in the following



- (a) (i) Graphite is often mixed in the polymer used for making the cathode electrode.

Suggest a reason for this.

[1]

To increase the electrical conductivity of the electrode
Or (good) conductor of electricity [1]

- (ii) Calculate the oxidation number of N in the cathode **before** discharge.

[1]

Oxidation number of N is +1 [1]

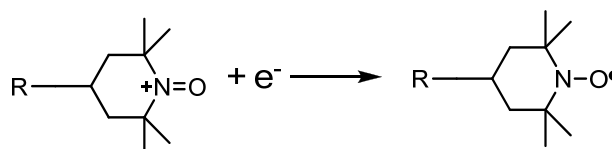
- (iii) The E_{cell} generated by the hybrid ORB/Li-ion battery under standard conditions is 2.17 V.

Use relevant E° value from the *Data Booklet* to calculate the electrode potential generated by the cathode half-cell.

[1]

$$\begin{aligned} E_{\text{cell}} &= E_{\text{red}} - E_{\text{ox}} \\ +2.17 &= E_{\text{red}} - (-3.04) \\ &= -0.87 \text{ V} \quad [1, \text{ with correct working and sign}] \end{aligned}$$

- (iv) During discharge, the following reaction occurs at the cathode.



where R represents the organic polymer cathode.

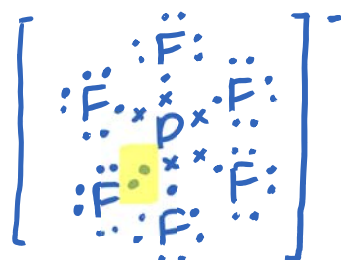
Write an equation for the overall process that occurs during discharge.

[1]



- (v) Draw the dot-and-cross diagram of the PF_6^- ion and state its shape.

[2]



[1, check for dative bond and overall charge]

Shape: **Octahedral** [1]

- (vi) Suggest whether LiPF_6 or LiF has a lower melting point. Explain your answer.

[2]

LiPF_6 . [1, conditional provided student considered LE)

PF_6^- has a **larger anionic radius** [✓] than F^- , giving rise to a **lower magnitude**

of LE [✓] (since $\text{L.E.} \propto \left| \frac{q_+ \times q_-}{r_+ + r_-} \right|$). Hence, LiPF_6 has a lower ionic bond strength

and less energy is required to overcome the ionic bond, giving rise to lower mp.

- (b) The hybrid ORB/Li-ion battery is a secondary battery, i.e., it is rechargeable.

During charging, 1.22 g of Li is regenerated from Li^+ ions at the cathode.

- (i) Calculate the amount of electrons required to form 1.22 g of Li. [1]

$$\text{Amount of Li formed} = 1.22 / 6.9 = 0.1768 \text{ mol}$$

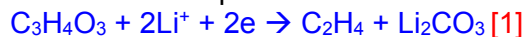


$$\text{Amount of electrons consumed} = 0.1768 = 0.177 \text{ mol [1].}$$

Besides the generation of Li, there is a competing side-reaction that occurs at the cathode.

In this side-reaction, ethylene carbonate, $\text{C}_3\text{H}_4\text{O}_3$ undergoes reduction in the presence of Li^+ ions to form ethene and lithium carbonate.

- (ii) Write the half-equation for the side-reaction occurring at the cathode. [1]



A current of 5.0 A is supplied over 2 hours during charging.

- (iii) Use the information given and your answer in (b)(i) to calculate the amount of electrons consumed by the side-reaction. [1]

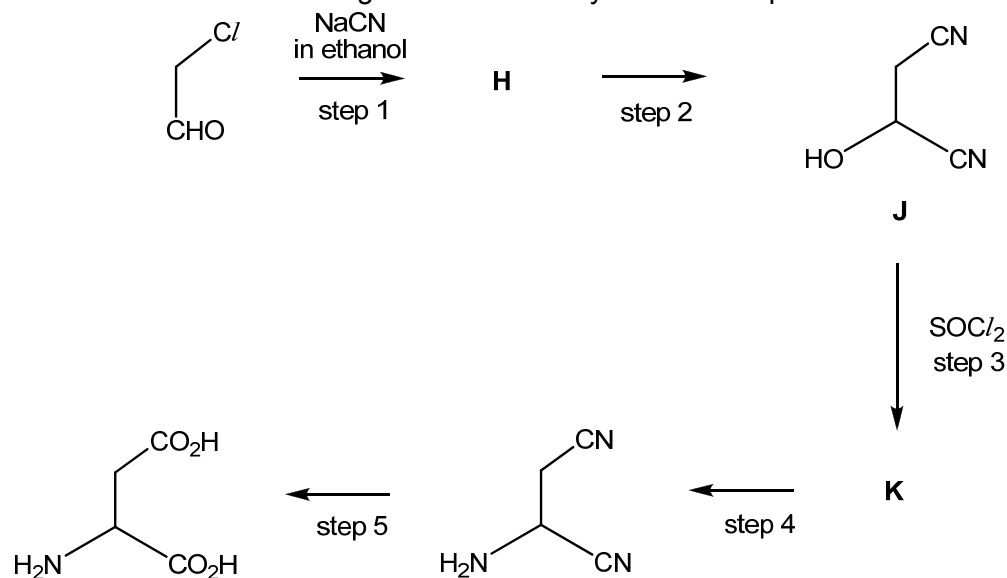
$$\text{Amount of electrons supplied} = (5.0) (2 \times 3600) / 96500 = 0.3731 \text{ mol [✓].}$$

$$\begin{aligned} \text{Hence, amount of electrons consumed by side-reaction} \\ &= 0.3731 - 0.1768 \\ &= 0.196 \text{ mol [✓, allow ecf from (i)].} \end{aligned}$$

- (iv) Suggest why the battery needs to be replaced after about 1000 charge-discharge cycles. [1]

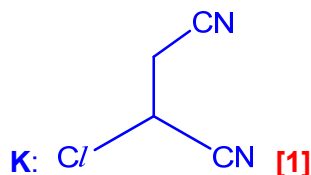
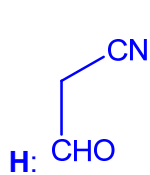
Some of the Li ions are depleted / not all the Li metal is regenerated during charging due to the side-reaction [1].

- (c) Chloroethanal is the starting material in the synthesis of aspartic acid.



- (i) Suggest structures for the intermediates **H** and **K**.

[2]



- (ii) Suggest reagents and conditions for step 2 and for step 4.

[2]

Step 2: HCN, trace NaOH(aq)/NaCN, cold **[1]**

Step 4: excess, concentrated ethanolic NH₃, heat in sealed tube **[1]**

- (iii) The reaction in step 2 produces sample **J**, which does not show optical activity. Explain the observation.

[2]

The nucleophile, CN⁻, has equal chances of attacking either above or below the plane of the electron-deficient carbonyl carbon in compound H **[1]**. Thus, the product mixture contains 50% of each of the enantiomer, forming a racemic mixture **[✓]**. Hence, the optical activity of the 2 enantiomers cancels out **[✓]**. Thus, sample **J** produced does not show optical activity.

[Total: 18]

Section B

Answer **one** question from this section

- 4 (a) For many compounds the enthalpy change of formation cannot be calculated directly. An indirect method based on enthalpy changes of combustion can be used.

The enthalpy change of combustion can be found by a calorimetry experiment in which the heat energy given off during combustion is used to heat a known mass of water and the temperature change recorded.

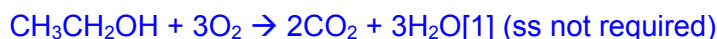
- (i) Define the term *standard enthalpy change of combustion*.

[1]

Standard enthalpy change of combustion, $\Delta H_c^\ominus$, of a compound is the enthalpy change (heat evolved) when 1 mole of the substance is completely burnt in oxygen under standard conditions of 298K and 1 bar. **[1]**

- (ii) Write the equation for the complete combustion of ethanol, CH₃CH₂OH.

[1]



In an experiment to determine the enthalpy change of combustion of ethanol, 0.23 g of ethanol was burned and the heat given off raised the temperature of 100 g of water by 16.3 °C.

- (iii) Calculate the heat energy change during the combustion of ethanol.

[1]

$$\begin{aligned} \text{Heat change} &= mc\Delta T \\ &= (100)(4.18)(16.3) \\ &= 6813.4 \end{aligned}$$

$$= 6810 \text{ J [1]}$$

(no mark awarded if unit given is wrong)

- (iv) Hence, calculate the enthalpy change on burning 1 mole of ethanol. [2]

$$\begin{aligned} \text{Amount of ethanol} &= 0.23 / (2 \times 12.0 + 6.0 + 16.0) \\ &= 5.00 \times 10^{-3} \text{ mol} \end{aligned}$$

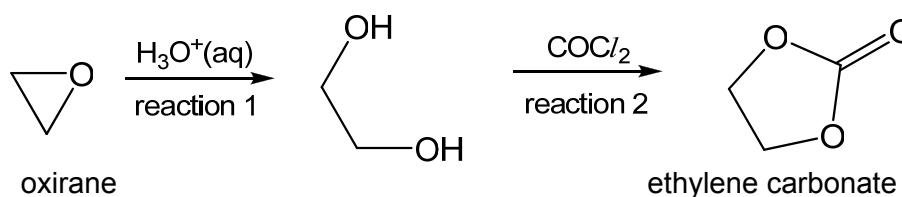
$$\begin{aligned} \text{Enthalpy change} &= - (6813.4 / 5.00 \times 10^{-3}) \text{ [1m; ECF from (iii)]} \\ &= - 1362680 \text{ J mol}^{-1} \\ &= - 1360 \text{ kJ mol}^{-1} \text{ [1m include unit \& sign]} \end{aligned}$$

- (v) Suggest one reason why the value for the enthalpy change of combustion of ethanol determined by a simple laboratory calorimetry experiment is likely to be lower than the true value. [1]

Heat loss to the surrounding. / Incomplete combustion/ Ethanol is volatile and evaporated./ Ethanol is impure. / Not all energy is absorbed by the water.
(Any answer 1 mark)

- (b) Epoxides are cyclic ethers commonly used in organic reactions.

Ethylene carbonate can be prepared from an epoxide, oxirane by the following reactions.



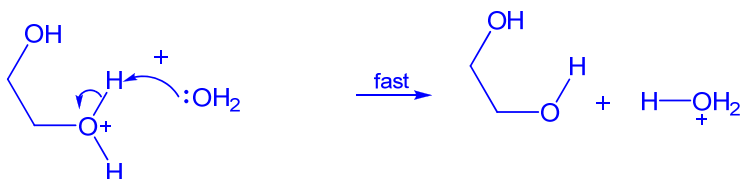
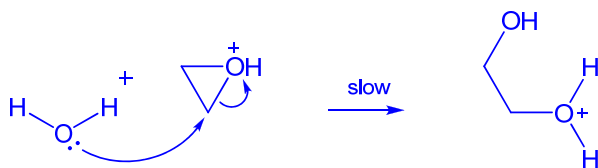
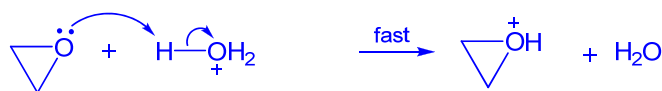
- (i) Suggest the type of reaction for reaction 2. [1]
Condensation/ nucleophilic acyl substitution [1]

- (ii) Reaction 1 is an acid-catalysed reaction that proceeds via a three-step mechanism:

- 1) Protonation of oxirane by H_3O^+ .
- 2) Ring opening of protonated oxirane due to nucleophilic attack by H_2O to yield an oxonium ion, $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}_2^+$.
- 3) Deprotonation of the oxonium ion to yield the product, with the regeneration of H_3O^+ .

Suggest the mechanism for Reaction 1.

[3]

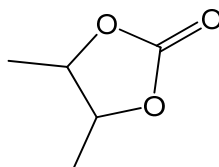


- ✓ - correct half-arrows and lone pairs for step 1
- ✓ - formation of H₂O in step 1
- ✓ - correct structure for protonated epoxide in step 1
- ✓ - correct half-arrows and lone pair for step 2
- ✓ - correct half-arrows and regeneration of H₃O⁺ for step 3

(fast & slow labels not marked for)

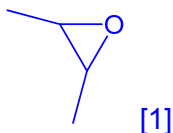
3m – all 5 ✓ ; 2m – 3 to 4 ✓ ; 1m – 2 ✓

Compound **L** can be synthesised from an epoxide in a similar manner as ethylene carbonate.



Compound **L**

- (iii) Draw the structure of the epoxide used for synthesising compound **L**. [1]



- (iv) Suggest why compound **L** is able to exhibit cis-trans isomerism. [1]
Restricted C-C bond rotation [1] due to ring strain/ rigidity of ring
 (and both C atoms of the ring is attached to 2 different groups or atoms)

- (c) A number of isomers with the formula Fe(H₂O)₆Cl₃ exist. Their general formula is [Fe(H₂O)_{6-n}Cl_n][Cl_{3-n}].nH₂O.

Each isomer contains a six co-ordinated Fe(III) ion in an octahedral complex. Water molecules not directly bonded with the Fe atom are held in the crystal lattice as water of crystallisation.

- (i) Similar to organic compounds, octahedral complexes can also exhibit stereoisomerism depending on the orientation of the ligands.

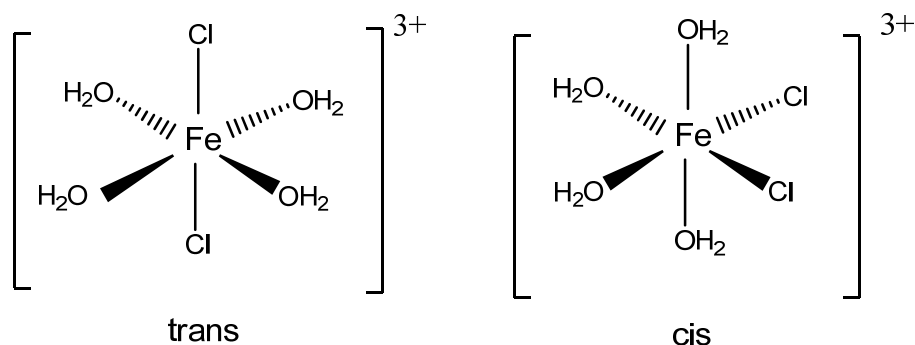
One such example will be the iron complex when n is 2. It can exist in two isomeric forms where only one of them has a dipole moment.

Name the type of isomerism shown by the complex

[1]

Cis- Trans isomerism

- (ii) Draw the structures of the two isomeric forms of the complex. [2]



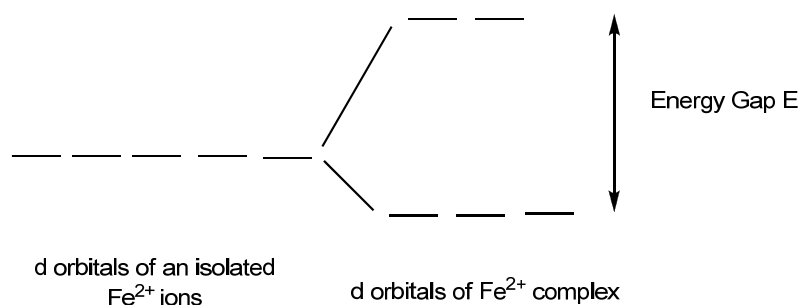
[1 mark each] (labelling of cis-trans not required)

- (iii) State which isomer has a dipole moment. Explain your answer. [2]

Cis isomer has a dipole moment. [1] (student need not mention "cis". Marks are awarded as long as student identify the correct isomer)

Cl⁻ on the same side of the complex/ overall dipole moment does not cancel out.[1]

- (iv) The following diagram shows how the d-orbitals are split in an octahedral environment.



When the H_2O ligand is changed to a Cl ligand, the Fe^{2+} ion changes the electronic configuration from a 'high spin' to a 'low spin' state.

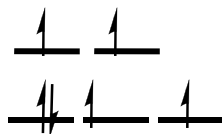
In a 'high spin' state, the electrons occupy all the d-orbitals singly, before starting to pair up in the lower energy d-orbitals.

In a 'low spin' state, the lower energy d-orbitals are filled first, by pairing up if necessary, before the higher energy d-orbitals are used.

Use diagrams like the one above to show the electronic distribution of a Fe^{2+} ion in a high spin state, and in a low spin state.

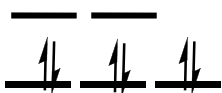
$\text{Fe}^{2+} : 3d^6$

'High spin' state



[1]

'Low spin' state



[1]

- (v) State and explain which ligand will result in a larger energy gap, E , between its d-orbitals.

[1]

(Complex with) Cl ligand [1]

Electrons will only pair up in the lower orbitals when the energy gap was greater than the interelectronic repulsion between the two electrons in the same orbital. (wtse) [1]

[Total: 20]

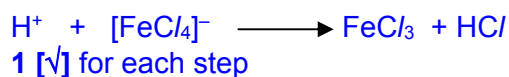
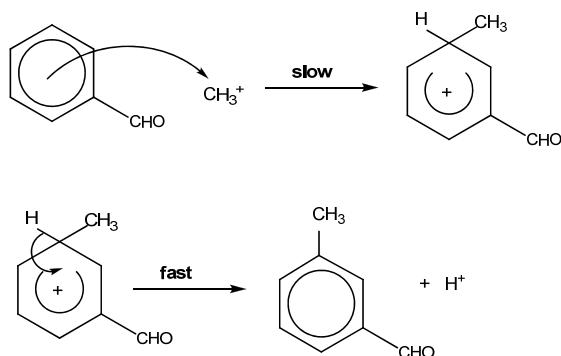
- 5 Carbonyl compounds are common in our everyday lives. They are mainly used as solvents, perfumes and flavouring agents or as intermediates in the manufacture of plastics and pharmaceuticals.

(a) The characteristic smell of cherries and fresh almonds is due to benzaldehyde.

- (i) Benzaldehyde can react with chloromethane to form 3-methylbenzaldehyde. Describe the mechanism for this reaction.

[3]

Electrophilic Substitution [1]



- (ii) Benzene can also react under a similar reaction with chloromethane.

State and explain whether benzene or benzaldehyde would react with chloromethane more readily.

[2]

Benzene would react **more readily** [✓] with chloromethane.

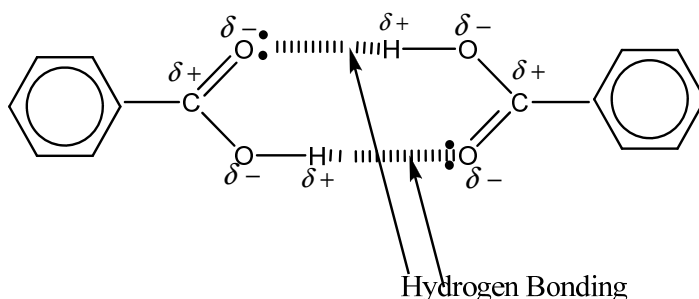
Benzaldehyde contains an **electron withdrawing CHO group** [✓] which **decreases the electron density of the benzene ring** [✓], making it **less susceptible to electrophilic attacks** [✓].

- (b) Benzaldehyde can also react with hot acidified dichromate(VI) to give benzoic acid. In benzene, benzoic acid associates to form dimers.



- (i) Draw a diagram to illustrate the bonding in the dimer.

[1]



All details included (dipoles, lone pair, hydrogen bonds) – [1]

- (ii) Predict and explain whether the dimerisation is favoured at a high or low temperature.

[2]

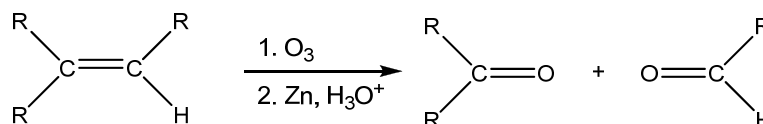
Low temperature [1, conditional]. At low temperature, the position of equilibrium shifts to the right to favour the **exothermic reaction [1]** to release heat.

- (iii) Suggest why the above equilibrium cannot be established in an aqueous medium.

In aqueous medium, **benzoic acid would form hydrogen bonds with the water molecules instead of forming a dimer, [1]**

[1]

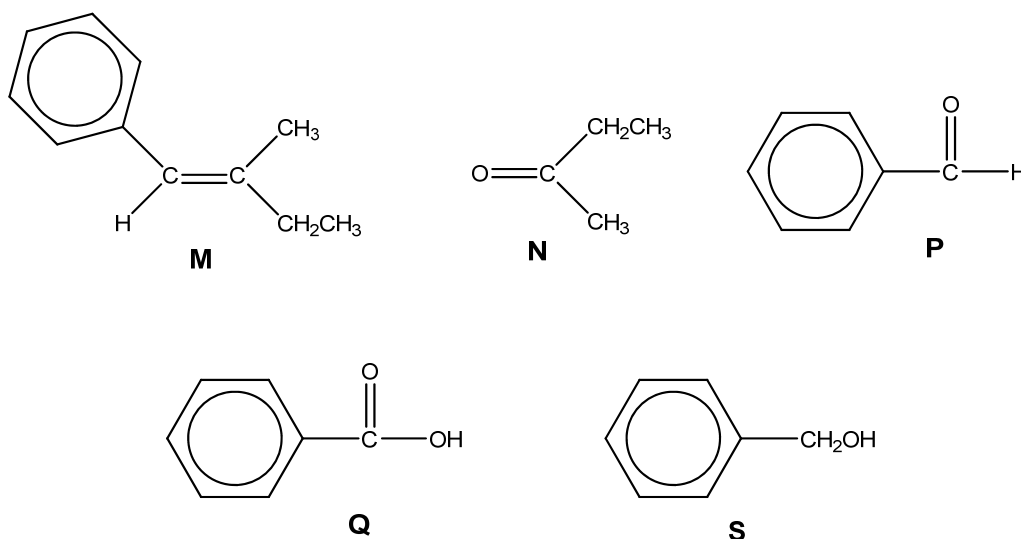
- (c) Carbonyl compounds can be prepared from alkenes via the ozonolysis reaction as shown below.



An alkene **M**, $\text{C}_{11}\text{H}_{14}$ was treated with O_3 , followed by Zn and H_3O^+ to give **N** $\text{C}_4\text{H}_8\text{O}$ and **P**, $\text{C}_7\text{H}_6\text{O}$. **N** gives a yellow precipitate with aqueous alkaline iodine while **P** gives a grey precipitate with ammonical silver nitrate solution. In the presence of OH^- , **P** undergoes a reaction to give **Q**, $\text{C}_7\text{H}_6\text{O}_2$ and **S**, $\text{C}_7\text{H}_8\text{O}$. Both **Q** and **S** react with Na metal, but only **Q** reacts with NaHCO_3 .

Suggest the structures of **M**, **N**, **P**, **Q** and **S**.

[5]



[1] each

- (d) **T**, **U**, **V** and **W** are four consecutive elements in the **fourth** period of the Periodic Table. (The letters are **not** the actual symbols of the elements.)

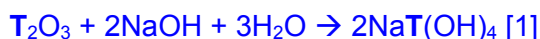
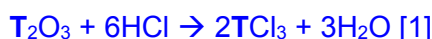
T is a soft, silvery metal with a melting point just above room temperature. Its amphoteric oxide, T_2O_3 , has a melting point of $1900\text{ }^{\circ}C$ and can be formed by heating **T** in oxygen.

W is a solid that can exist as several allotropes, most of which contain W_8 molecules. **W** burns in air to form WO_2 and WO_3 , which dissolves in water to form an acidic solution. The acidic solutions react with sodium hydroxide to form the salt Na_2WO_3 and Na_2WO_4 respectively.

- (i) Suggest the identities of **T** and **W**. [2]

T = Ga [1] **W** = Se [1]

- (ii) Write equations for the reactions of T_2O_3 with
- hydrochloric acid,
 - sodium hydroxide
- [2]

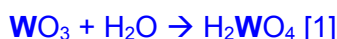


(in writing the equations, students can replace **T** with Ga)

- (iii) Suggest the structure in T_2O_3 . [1]

Giant Ionic Structure [1]

- (iv) Write an equation for the formation of the acidic solution when WO_3 dissolves in water. [1]



(in writing the equation, students can replace **W** with Se)

[Total: 20]